

Supplementary Materials for Qualifying Paper II

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A List of experimental stimuli

A.1 Exp. 1 - Full Information

Target

We want to buy Thomas some more sweaters ____ perhaps a cardigan.

We want him orange ____ pink.

If she wants us to read the book ____ like take the shoes off.

I want a lady hairdresser ____ a man hairdresser.

I think traffic was good on the motorway ____ they set off earlier.

I think Anna's heard that story once ____ twice.

I think it's mostly brown ____ black.

I think that's grey ____ silver.

Would you like for example um a tin of strawberries ____ a tin of mixed fruit?

Do you want orange juice ____ milk?

Do you want mommy to pour it ____ Lily to pour it?

Do you wanna read that book with Mom ____ just look at it yourself?

Do you think it's summer ____ winter there?

Do you think he's gonna face forwards ____ backwards?

Do you think it's Linda ____ Brian?

Do you think it's a fly ____ a bee?

Grampy wanted to buy you a new bird yesterday ____ and a turtle.

You want me ____ you to do this puzzle.

Okay so you wanted to see Sully ____ Boo first.

I wanna come out ____ play.

I think it's supposed to be a farmhouse ____ there's a fence.

I think Jessica's gonna go ____ get something.

I think I'll get some glue ____ some paper.

I think it is pasta ____ meatballs.

Do you wanna sit on the stool ____ read the book?

Do you wanna lie down on the couch ____ put your head on the pillow?

Would you like some raisins, raisins ____ sultanas?

Do you wanna go see the cows ____ the horsies?

Do you think it's gonna be about a bicycle or a cat ____ a dog?

Do you think it was Granddad ____ Grandma?

Do you know where Daddy ____ Oma went?

Do you think the train has broken down ____ they've all had to get out and walk?

Distractors (response options in brackets)

I hope that the books ____ on the table. (are, to be)

I remember that the two boys ____ sitting on that bench. (to be, are)

I know that James and Lois ____ going to be late. (are, to be)

Emma thinks that the bears ____ too picky. (to be, are)

Do you think that James and Sue ____ going to come? (are, to be)

Did you know that the books ____ going to arrive tomorrow?	(to be, are)
Will Mary realize that the bears ____ dancing?	(are, to be)
Will anyone know that the prizes ____ hidden here?	(to be, are)
Eloise wants there ____ at least three choices.	(are, to be)
Susie needs Em and James ____ on time.	(to be, are)
I hope for the boys ____ the best.	(are, to be)
I'd like the presents ____ opened tomorrow.	(to be, are)
Would Jamie like three or four books ____ on the table?	(are, to be)
Do you want the noodles ____ warmed up?	(to be, are)
Does Susan want us ____ early tomorrow?	(are, to be)
Do the books need ____ put back anywhere special?	(to be, are)

A.2 Exp. 2 - Partial Information (Jabberwocky sentences)

Target

We want to plew Snork some sloyish flosps ____ a masp.

We want him woonky ____ snaffy .

She wants us to snam the flosps ____ woonk the masps off.

I want a snaf masp ____ a flosp masp.

I think masp was sloyish on the flosp ____ they plewed off sterkly.

I think Snork has snaffed that flosp sleekily ____ sloyly.

I think it is sleekily trinky ____ sloyish.

I think that is reemby ____ clangish.

Would you like a snaf of slents ____ a masp of reemby flosps?

Do you want snaffish masp ____ fosp?

Do you want Snork to plew it ____ Slent to plew it?

Do you wanna snam that fosp with Slent ____ sterkly woonk at it yourself?

Do you think it's snaf ____ masp there?

Do you think he's gonna snam fosp ____ masp?

Do you think it's Slent ____ Snork?

Do you think it's a snaf ____ a masp?

Slent wanted to plew you a clangish snaf ____ a masp.

You want me ____ you to do this fosp.

Okay, so you wanted to snam Snork ____ Slent sterkly.

I wanna snam out ____ plew.

I think it's snammed to be a fosp ____ there's a masp.

I think Snork's gonna plew ____ snam snaf.

I think I'll plew some snaf ____ some fosp.

I think it's snaf ____ masp.

Do you wanna whuk on the masp ____ snam the fosp?

Do you wanna whuk down on the snam ____ woonk your fosp on the masp?

Would you like some masps, masps ____ flosps?

Do you wanna plew the masps ____ the flosps?

Do you think it's gonna be about a fosp ____ a masp?

Do you think it was Slent ____ Snork?

Do you know where Slent ____ Snork whukked?

Do you think the snaf has plewed down ____ they've all had to woonk out and snam?

Distractors (response options in brackets)

I hope that the masps ____ on the flosp. (are, to be)

I remember that the two masps ____ whukking on that snaf. (to be, are)

I know that Slent and Snork ____ plewing to be barkish. (are, to be)

Slent thinks that the masps ____ too sloyish. (to be, are)

Do you think that Slent and Snork ____ woonking to snam? (are, to be)

Did you know that the masps ____ woonking to snam barkly? (to be, are)

Will Snork realize that the flosps ____ woonking? (are, to be)

Will Slent know that the snafs ____ plewed barkly? (to be, are)

Slent wants there ____ at least three snafs. (are, to be)

Snork needs Slent and Flent ____ on masp. (to be, are)

I hope for the flosps ____ the sloyest. (are, to be)

I'd like the snafs ____ snammed clangish. (to be, are)

Would slent like three or four slents ____ on the snaf? (are, to be)

Do you want the slents ____ whukked up? (to be, are)

Does Slent want us ____ reemby masp? (are, to be)

Do the snafs need ____ snammed back sleekly? (to be, are)

B Additional corpus investigations

In addition to the cues mentioned in the main text, we looked at a set of other cues meant to track additional syntactic properties of uses of disjunction, alongside other loose indicators of possibility/uncertainty that might inform the learner.

B.1 Coding additional cues

B.1.1 Global corpus

Negation. For each token, we code whether it was an occurrence of a set of negative expressions. For each word we coded it as instantiating one of the following negative expressions (*not*, *no*, *noone*, *none*, *never*, *nobody*), or as non-negative. This was done in two different ways: first, using the stem value of a given token, we checked for matches with each of the negative expressions listed above. Then, to catch instances of clitic negation (*n't*), we checked for occurrences of the value ‘neg’ in a word’s clitic code, or for occurrences of *n't* in a word’s gloss. These were grouped together with *not* for our analyses.

Modals. For each token, we coded whether it was one of the modal adverbs or auxiliaries listed in Table 1. As with attitudes, this was done by looking at the stem value for a given word, and only including items whose part of speech was marked as ‘mod’. Later frequencies calculated on this basis exclude all tokens missing a value for stem. Table 1 shows the full list of modals. We further coded modals on the basis of force, and predominantly used this broad force distinction for our subsequent analyses.

B.1.2 Hand-annotated corpus

Syntactic category of coordinands. This variable was used to draw a broad distinction between nominal, verbal and sentential coordination, with levels N, V, S, and OTHER.

	possibility	necessity
AUX	<i>can</i> <i>could</i> <i>may</i> <i>might</i>	<i>have to</i> <i>should</i> <i>must</i> <i>ought</i> <i>got to</i>
adverbs ¹	<i>maybe</i> <i>possibly</i> <i>probably</i>	<i>certainly</i> <i>absolutely</i> <i>definitely</i>

Table 1: Modals extracted and coded by syntactic category and modal force

(1) Nominal coordinations (N):

- a. Would you like to make a round moon or a crescent shaped moon?
- b. Cookie and milk is a snack.

(2) Verbal coordinations (V):

- a. Does it stay on or go off?
- b. You gonna climb out and fall on your head.

(3) Sentential coordinations (S):

- a. Did you bring her inside, or is she still in the car?
- b. They're gonna make you thirsty, and then you'll drink your water.

Negation. We coded for the presence of sentential negation, and negative expressions. We make a broad distinction between sentential negation and other negation markers. Sentential negations appears in two forms, clitic negation (as in ‘don’t’) and as a discrete negative adverb, ‘not’. We coded and marked separately the appearance of other negative expressions, of which there were relatively few in the corpus: we only found occurrences of ‘no’ and ‘never’.

- (4) a. I didn’t want it to melt on the floor and make a mess.

possibility	necessity
<i>can</i>	<i>must</i>
<i>might</i>	<i>should</i>
<i>could</i>	<i>have to</i>
<i>maybe</i>	<i>got to</i>
<i>possibly</i>	<i>had better</i>
<i>probably</i>	

Figure 1: Modals in the hand-annotated corpus

- b. No Marmite, and no spreading cheese.
- c. Oh, I don't know if the wrench will stand up there or not.

Modals. We coded for the presence of modal verbs or lexical modals. The full list of modal expressions found in the corpus is given in Fig. 1. Looking at a finer level of granularity than the mere presence of modal marking, we were interested in the modal force distinction as well. In Fig. 1 the modals are listed according to their force.

- (5) a. When you go to Stockport you can see the bridge very high up can't you?
and you can see trains going across the top of the bridge.
- b. Perhaps rubber or vinyl.
- c. We've got to put a star or a fairy on top.

Past tense marking. For tense, we make a broad distinction between past and non-past marking on tensed clauses, because this is a relatively transparent and visible measure. The motivation comes from the fact that past tense descriptions are generally settled in a way that future oriented expressions are not. Past-marked expressions tend to be claims about settled facts of the world. Drawing the distinction between past and non-past expressions is most straightforward to code — the temporal orientation of present tense marked expressions without an explicit future auxiliary is

generally mixed. Some expressions can have a present orientation, in which case they will share the property of making claims about settled facts with past marked expressions, while other present tense expressions in English can have futurate interpretations. This allows, albeit somewhat indirectly, to assess the extent to which disjunction and conjunction differ in interacting with settled, world-directed claims.

(6) Past-marked expressions:

- a. There were two big gorillas and a baby gorilla swinging in the trees
- b. Um, I can't decide if she liked, or didn't like the squishy ones.
- c. Did Auntie Linda have any paper chains up, or were those just streamers?

(7) Non-past expressions:

- a. A frog, you're looking for a frog or a watermelon.
- b. I think it's either a dog or a wolf.
- c. Let's put it on the floor and watch it go.

B.2 Results across corpora

Syntactic categories of coordinands. Fig. 2 shows the distribution of syntactic categories of coordinands. We see that disjunction occurs more often with nominal coordinands than conjunction: 43% of disjunctions were nominal, compared with 26% of conjunctions. The reverse pattern holds for sentential or verbal categories, both of which make up a larger share of conjunctions than disjunctions.

Negation. In the global corpus, the overall frequency of any kind of negation in parental speech was 31.5 tokens per 1000. Fig. 3 shows the frequencies of particular kinds negative expressions.

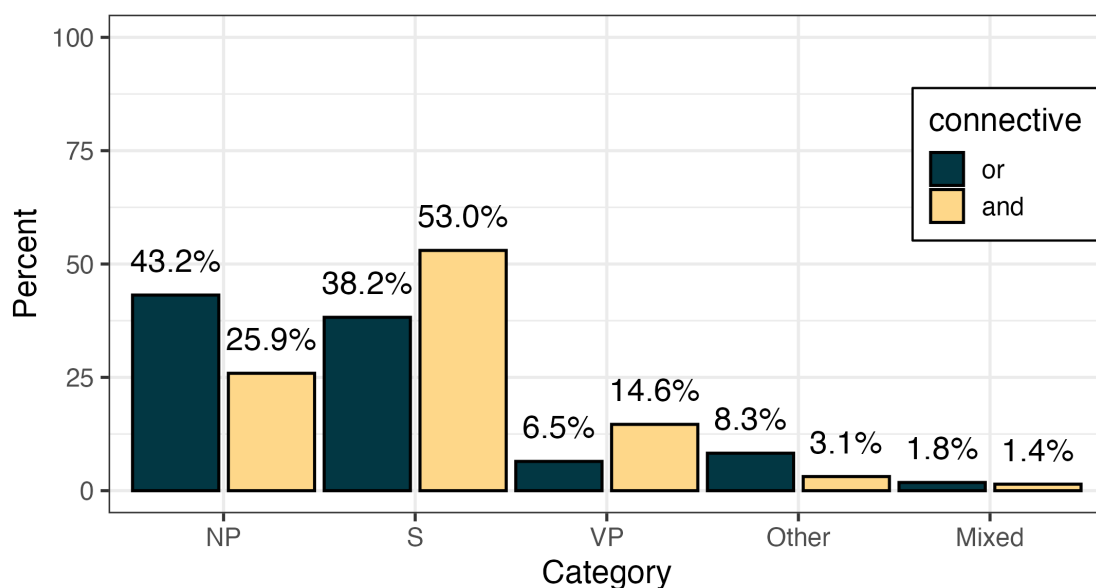


Figure 2: Percentage of syntactic category of coordinands by connective in the hand-annotated corpus

We see that the predominant form of negation is *not*, either alone or contracted with an auxiliary. The negative expression *no* made up another 25%, at 8.0 occurrences per 1000 tokens, and all other negative expressions (*never*, *nobody*, *none* and *no one* made up the remaining occurrences, with low individual frequencies).

Fig. 4a shows the relative frequencies of *and* and *or* both in and outside the context of negative expression. We see that in fact, both disjunction and conjunction become less frequent in the context of negation. The frequency of *or* drops from 1.9 outside of negation to 1.6 in a negative context; *and* drops from 19.0 to 14.1.

Fig. 4b shows the percentage of utterances with negation scoping over a connective in the hand-annotated corpus. Here, we observe a contrast between disjunction and conjunction: Negation occurs about 6% of the time with disjunction, vs. 2% of the time with conjunction.

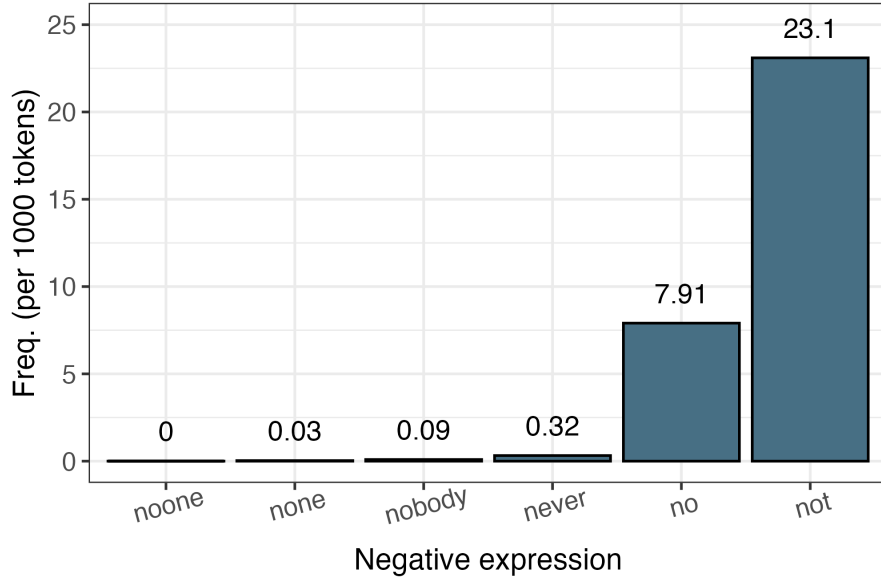
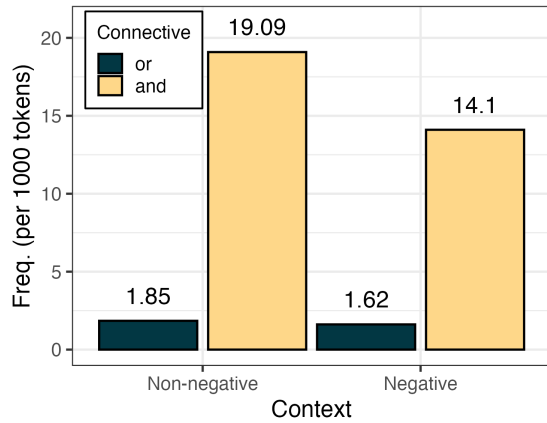
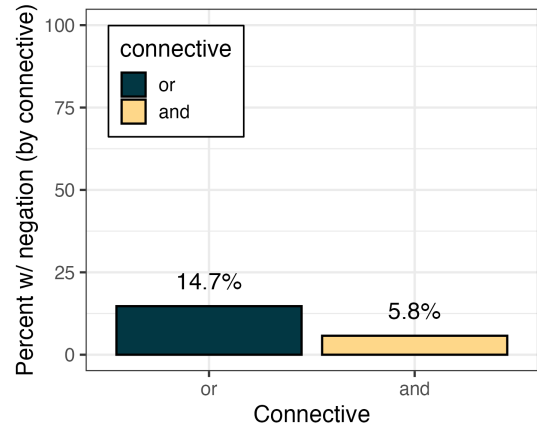


Figure 3: Forms of negation and their relative frequencies per 1000 words in the input (global corpus data)



(a) Connective frequencies inside/outside the context of a negative expression in the global corpus



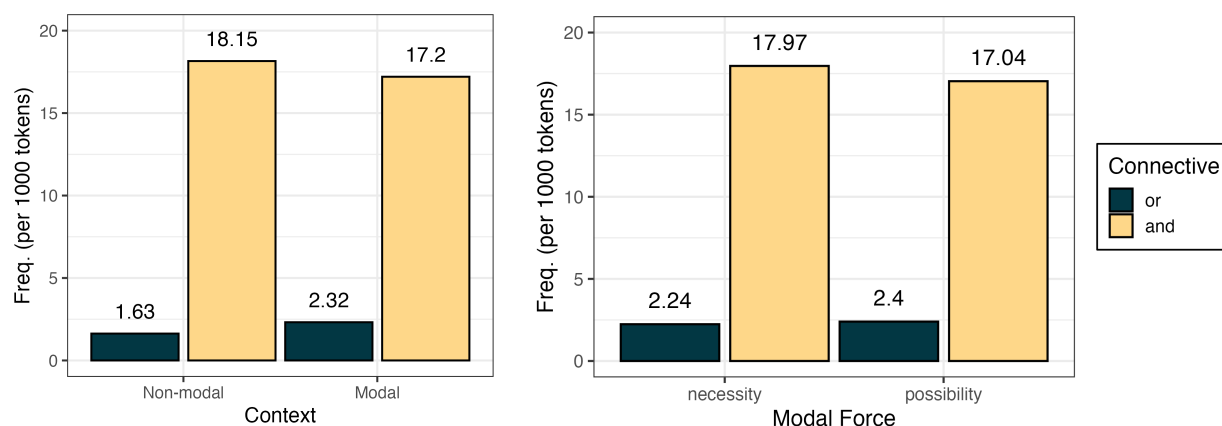
(b) Percentage of utterances containing negation, per connective, in the hand-annotated corpus

Figure 4: Negation in the global corpus (a) and the hand-annotated corpus (b).

Modals. In the global corpus we see that the overall prevalence of modal items is 16.0 per 1000 tokens in the input. More narrowly, we see a frequency of 3.5 per 1000 for necessity modals and 11.0 per 1000 tokens for possibility modals. By far the most frequent modal verb in the input is *can*, at 8.1 occurrences per 1000 tokens. The next two most frequent modals are *have to* at 1.8

occurrences per 1000 tokens and *could* at 1.3 occurrences per 1000 tokens.

Fig. 5a shows the overall frequencies of *and* and *or* in the context of any modal (regardless of the force). The word *or* appears more frequently than baseline, at 2.3 occurrences per 1000, vs. 17.2 occurrences per 1000 for *and*. Fig. 5b further shows the relative frequencies of *and* and *or* broken down by modal force (possibility, necessity). We see comparable behavior regardless of modal force.



(a) By general co-occurrence with modal- (b) Frequencies by context defined by co-occurrence with particular modal force

Figure 5: Connective frequencies in various modal contexts in the global corpus

Fig. 6 further shows the percentages of utterances in the hand-annotated corpus that contained at least one necessity or possibility modal, broken down by connective. Differences here appear quite subtle, with possibility modals being slightly more common among disjunctive utterances, relative to conjunctive utterances. 10.3 % of utterances with disjunction had at least one possibility modal, vs. 7.4% of utterances with conjunction.

For necessity contexts, the relative frequencies of *or* and *and* are 1.9 and 21 respectively. **Tense.** Fig. 7 shows the overall proportions of utterances by temporal orientation in the hand-annotated corpus. Past tense marking is more likely in conjunctive utterances (17 % of conjunc-

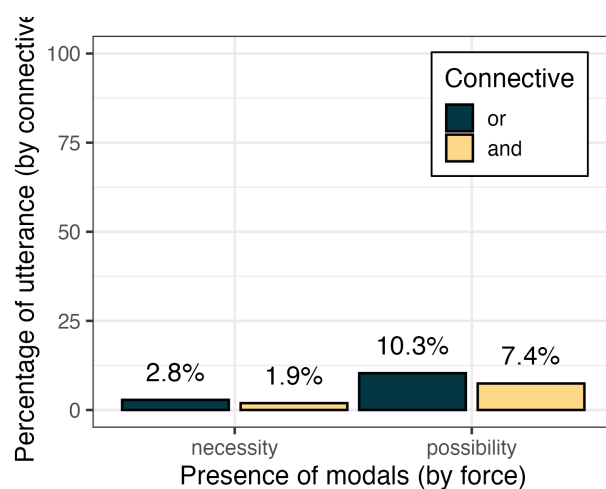


Figure 6: Percentages of utterances containing possibility or necessity modals in the hand-annotated corpus

tions) vs. disjunction (8.9% of disjunction).

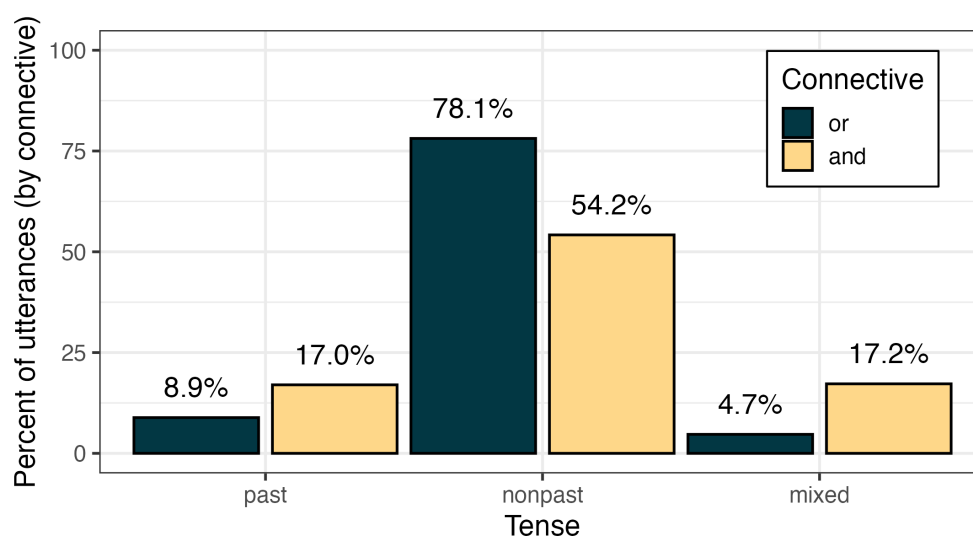


Figure 7: Percentages of past-marked, non-past and mixed tense utterances by connective, within all tensed utterances in the hand-annotated corpus